

Executive Platform Architecture, Ingestion Governance & Security Protocol

Confidential Briefing for Senior Leadership · Target Version: v14 Production Standard · IST Timezone

1. Executive Summary & Core Purpose

The **Bambinos Marketing Intelligence Suite** is a unified, dual-truth analytics platform engineered to provide 100% visibility into acquisition unit economics, lead maturation cohorts, financial ROAS, and campaign diagnostics. It eliminates conversion ambiguity by bridging two separate sources of truth:

- **Meta Ads Platform Delivery (via Windsor.ai):** Ingests granular advertising data (Spend, Impressions, Link Clicks, CPM, CPC, Landing Page Views, and Meta conversion actions) broken down by Campaign, Ad Set, Ad, Placement, and Device OS.
- **Internal Business Outcomes (via Metabase / Database):** Tracks verified CRM records (Unique Contact Submissions, Demos Booked, Demos Attended, Paying Enrolments, and Governed New-Business Revenue).

2. The 5 Canonical Intelligence Modules

Module	Executive Scope	Key Metrics & Decision Value
1. Company Level ROAS	Calendar-date financial reconciliation by invoice/payment date.	Marketing Spend, New Revenue, Renewal Revenue, Company ROAS, Cost per Demo Booked, Revenue per Demo Booked.
2. Cohort Performance Explorer	Permanent lead-capture cohort tracking across D0–D30 maturity windows.	Fixed D0 acquisition investment vs moving mature conversions, attendance %, ARPU, and multi-dimensional slicing.
3. Cohort ROAS Progression	Maturation Heatmap Matrix visualizing lead capture date vs outcome window.	Dynamic HSL color grading for ROAS, Lead Conversion %, and Demos Attended across D0 through Till Date.
4. Campaign Meta Diagnosis	Comparative diagnostic engine (Performance Window vs Comparison Window).	6-stage funnel volume audit + 5-driver multiplicative cost pressure decomposition (CPM, CTR, Landing %, Lead %, Booking %).
5. Meta Ads Manager Data	Ad delivery facts and creative/device distribution directly from Meta.	Impression Device OS (iOS vs Android vs Desktop), placement efficiency, and independent booking event selection.

3. Third-Party Ingestion Governance & API Cost Control Plan

To guarantee **zero runaway API costs and minimal token/credit consumption**, Windsor.ai integration operates under strict architectural constraints:

Cost / Token Guardrail	Implementation Mechanism	Financial & Operational Benefit
1. Zero On-Demand User API Calls	The dashboard frontend & API endpoints read 100% from precomputed local cache (JSON/Redis).	Users switching tabs, filters, or date ranges generate 0 external API calls and 0 token cost .
2. Single Daily Batch Ingestion	Scheduled once per 24 hours at 08:00 IST (02:30 UTC) covering the previous completed day.	Eliminates continuous polling, erratic scrapers, and background quota drain.
3. Incremental Delta Sync (Active Window)	Historical months (>30 days) are frozen as immutable ledgers; only the moving 30-day window is queried.	Reduces payload transfer volume by over 85% and avoids repetitive re-fetching of historical facts.
4. Consolidated Field Selection	All necessary metrics, conversion actions, placements, and devices are pulled in a single batch query.	Prevents redundant multi-query breakdowns for the same date/account grain.

5. Hard Circuit Breakers & Quota Caps	Maximum 3 retries with exponential backoff; hard cap of max 5 API requests per day.	Prevents runaway infinite retry loops during network or third-party outages.
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4. Data Security, Privacy & Anti-Breach Safeguards

Protecting company proprietary data, customer privacy, and preventing unauthorized third-party access or data leakage is paramount. The system enforces the following **zero-trust security policies**:

- **Zero External Data Exposure (Air-Gapped Ingestion)**: Third-party connectors (Windsor.ai, Meta API) are strictly *read-only* and *one-way*. No internal customer records, payment logs, or database credentials are ever sent to Windsor or any external third party.
- **PII Protection & Hash Masking**: Raw lead phone numbers and parent contact information are restricted to authorized admissions staff under role-based access control (RBAC). In analytical and reporting aggregates, identifiers are masked or one-way tokenized.
- **Server-Side Secret Isolation**: All API tokens (WINDSOR_API_KEY, METABASE_API_KEY, DB passwords) reside exclusively in encrypted server environment variables. They are never transmitted to the browser client or exposed in JavaScript.
- **Masked Error Logging & Sanitization**: The ingestion and analytical engine deliberately filters out raw HTTP request bodies, headers, and API responses from logs to prevent accidental credential leakage during service exceptions.
- **Staging Snapshot Integrity & Gating**: Daily data is ingested into an isolated staging sandbox first. It requires an automated checksum receipt (success.json) verifying row integrity, positive spend, and non-empty conversion columns before promotion to production.

5. Strict Calculation Contracts & Campaign Governance

The Cardinal Law: 'Never Average Ratios'. All unit ratios (ROAS, Cost/Demo, Attendance %, Lead Conversion %) divide filtered aggregated sums after all slicers are applied:

- **Cost per Demo Booked** = $\$ \sum \text{Spend} / \sum \text{Demos Booked} \$$ (Sum-first, zero bookings displays an em dash).
- **Revenue per Demo Booked** = $\$ \sum \text{New Revenue} / \sum \text{Demos Booked} \$$ (Governed new_revenue).
- **Anti-Double-Counting for Meta Events**: CompleteRegistration and StartTrial are independent booking signals and are never summed together.
- **Campaign Governance (campaign-tags.csv)**: Standardized registry matching names/aliases to canonical Display Name, Course, Market, and Channel. Unmatched campaigns strictly become Unmapped.

6. Current Progress & Project Status

Phase	Deliverable & Scope	Status
Phase 1	Metric Contracts, Booking Economics & Gating Engine	✓ COMPLETED
Phase 2	FastAPI Modular Backend (/revenue, /analytics, /operations, /catalog)	✓ COMPLETED
Phase 3	Frontend UI with 5 Canonical Tabs, Heatmap Matrix & Driver Charts	✓ COMPLETED
Phase 4	Automated Windsor Ingestion, Token Control & Daily 08:00 IST Workflow	■ IN EXECUTION
Phase 5	Reconciliation & Full Production Deployment	Pending Phase 4